Chapter 8

Quadrilaterals

Unit 3 — Quadrilaterals and Circles

McGraw Hill Glencoe Geometry Texas, 2015

Quadrilateral Hierarchy

```
         Quadrilateral
               |
         Parallelogram
         /      |      \
   Rectangle   Rhombus   (both)
         \      |      /
           Square
```

Also: **Trapezoid** (exactly one pair of parallel sides)

Kite (two pairs of consecutive congruent sides)

Key Vocabulary

Term	Definition
Parallelogram	Quadrilateral with two pairs of parallel sides
Rectangle	Parallelogram with four right angles
Rhombus	Parallelogram with four congruent sides
Square	Parallelogram with four right angles and four congruent sides
Trapezoid	Exactly one pair of parallel sides
Isosceles trapezoid	Trapezoid with congruent legs
Kite	Two pairs of consecutive congruent sides

Theorem 8-1: Parallelogram — Opposite Sides

Opposite sides of a **parallelogram** are **congruent**.

$$\overline{AB} \cong \overline{CD}, \quad \overline{BC} \cong \overline{AD}$$

Theorem 8-2: Parallelogram — Opposite Angles

> Opposite angles of a **parallelogram** are **congruent**.

$$\angle A \cong \angle C, \quad \angle B \cong \angle D$$

Theorem 8-3: Parallelogram — Consecutive Angles

> Consecutive angles in a **parallelogram** are **supplementary**.

$$m\angle A + m\angle B = 180^\circ$$

Theorem 8-4: Parallelogram — Diagonals Bisect

> The **diagonals** of a parallelogram **bisect** each other.

$$AE = CE \quad \text{and} \quad BE = DE$$

(where E is the intersection of the diagonals)

Proving a Parallelogram

Condition	Sufficient to prove $ABCD$ is a parallelogram
Thm 8-5	Both pairs of opposite sides congruent
Thm 8-6	Both pairs of opposite angles congruent
Thm 8-7	Diagonals bisect each other
Thm 8-8	One pair of sides both $\parallel$ and $\cong$

Theorem 8-9: Rectangle — Diagonals Congruent

> The **diagonals** of a rectangle are **congruent**.

$$\overline{AC} \cong \overline{BD}$$

Theorem 8-10: Rhombus — Diagonals Perpendicular

The **diagonals** of a rhombus are **perpendicular**.

$$\overline{AC} \perp \overline{BD}$$

Theorem 8-11: Rhombus — Diagonals Bisect Angles

> Each **diagonal** of a rhombus **bisects** a pair of opposite angles.

$$\overrightarrow{AC} \text{ bisects } \angle A \text{ and } \angle C$$

Properties Summary

Property	Parallelogram	Rectangle	Rhombus	Square
Opp. sides $\parallel$	✓	✓	✓	✓
Opp. sides $\cong$	✓	✓	✓	✓
Opp. $\angle$s $\cong$	✓	✓	✓	✓
Diag. bisect	✓	✓	✓	✓
Diag. $\cong$	—	✓	—	✓
Diag. $\perp$	—	—	✓	✓
All $\angle$s $= 90^\circ$	—	✓	—	✓
All sides $\cong$	—	—	✓	✓

Theorem 8-14: Isosceles Trapezoid — Base Angles

Both pairs of **base angles** of an isosceles trapezoid are **congruent**.

$$\angle A \cong \angle B \quad \text{and} \quad \angle C \cong \angle D$$

Theorem 8-15: Isosceles Trapezoid — Diagonals

> The **diagonals** of an isosceles trapezoid are **congruent**.

$$\overline{AC} \cong \overline{BD}$$

Theorem 8-16: Trapezoid Midsegment

> The **midsegment** of a trapezoid is parallel to the bases and its length equals half the sum of the bases.

$$MN \parallel AB,\ MN \parallel CD \qquad MN = \frac{AB + CD}{2}$$

Example — Trapezoid Midsegment

A trapezoid has bases of length 8 and 14. Find the midsegment.

$$MN = \frac{8 + 14}{2} = \frac{22}{2} = 11$$

Chapter 8 — Theorem Summary

#	Name	Key Idea
Thm 8-1	Parallelogram — Sides	Opp. sides $\cong$
Thm 8-2	Parallelogram — Angles	Opp. $\angle$s $\cong$
Thm 8-3	Parallelogram — Consec.	Consec. $\angle$s supp.
Thm 8-4	Parallelogram — Diag.	Diagonals bisect each other
Thm 8-9	Rectangle — Diag.	Diagonals $\cong$
Thm 8-10	Rhombus — Diag.	Diagonals $\perp$
Thm 8-11	Rhombus — Bisect	Diagonals bisect vertex $\angle$s
Thm 8-14	Isos. Trap. — Angles	Base $\angle$s $\cong$
Thm 8-15	Isos. Trap. — Diag.	Diagonals $\cong$
Thm 8-16	Trap. Midsegment	$MN = \frac{1}{2}(b_1 + b_2)$